

EDA-Data Inspection and Analysis

Code:

```
import pandas as pd

df = pd.read_csv('test_Y3wMUE5_7gLdaTN.csv')

print("--- df.head() ---")
print(df.head())

print("\n--- df.tail() ---")
print(df.tail())

print("\n--- df.info() ---")
print(df.info())

print("\n--- df.describe() ---")
print(df.describe())

print("\n--- df.columns ---")
print(df.columns)

categorical_cols = ['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'Property_Area']

for col in categorical_cols:
    unique_vals = df[col].unique()
    print(f"\nUnique values in '{col}': {unique_vals}")

df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'], errors='coerce').astype('Int64')

print("\nValue counts in 'Dependents' after conversion:")
print(df['Dependents'].value_counts(dropna=False))

cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']

for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    # Mode returns a Series, so we use .values
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

Output:

```
--- df.head() ---
   Loan_ID Gender Married ... Loan_Amount_Term Credit_History Property_Area
0  LP001000 Male      No ...             360.0             1.0          Rural
1  LP001001  NaN      Yes ...             360.0             1.0          Urban
2  LP001002 Male      Yes ...             480.0             1.0          Urban
3  LP001003 Male      Yes ...              12.0             1.0          Urban
4  LP001004 Male      Yes ...             300.0             0.0          Urban

[5 rows x 12 columns]

--- df.tail() ---
   Loan_ID Gender Married ... Loan_Amount_Term Credit_History Property_Area
362 LP001362 Male      Yes ...              84.0             1.0          Urban
363 LP001363 Male      Yes ...             180.0             0.0        Semiurban
364 LP001364 Female    No ...              84.0             1.0          Rural
365 LP001365 Male      No ...             240.0             1.0          Urban
366 LP001366  NaN      Yes ...             180.0             1.0        Semiurban

[5 rows x 12 columns]

--- df.info() ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Loan_ID               367 non-null    object
 1   Gender                350 non-null    object
 2   Married               343 non-null    object
 3   Dependents            356 non-null    object
 4   Education             367 non-null    object
 5   Self_Employed         345 non-null    object
 6   ApplicantIncome       367 non-null    int64
 7   CoapplicantIncome     367 non-null    int64
 8   LoanAmount            366 non-null    float64
 9   Loan_Amount_Term      330 non-null    float64
10   Credit_History         333 non-null    float64
11   Property_Area         367 non-null    object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None

--- df.describe() ---

```

	ApplicantIncome	CoapplicantIncome	...	Loan_Amount_Term	Credit_History
count	367.000000	367.000000	...	330.000000	333.000000
mean	38698.918256	19750.910082	...	197.309091	0.759760
std	23702.016502	11756.824989	...	148.075173	0.427872
min	312.000000	141.000000	...	12.000000	0.000000
25%	18235.000000	9478.000000	...	60.000000	1.000000
50%	36885.000000	18710.000000	...	180.000000	1.000000
75%	54498.500000	30101.000000	...	300.000000	1.000000

```
Median: 36885.00
Mode: [15529]
Range: 80635.00
Variance: 561785586.28
Standard Deviation: 23702.02

Statistics for 'CoapplicantIncome':
Mean: 19750.91
Median: 18710.00
Mode: [ 141  155  193  587  645  726  778 1089 1125 1141 1283 1492
1529 1875 1958 1998 2044 2261 2282 2307 2454 2591 2597 2659
2800 3112 3134 3267 3311 3321 3744 3760 3828 4033 4079 4088
4249 4378 4492 4493 4648 4658 4912 5075 5374 5410 5433 5463
5482 5808 5827 5889 5906 6031 6142 6317 6370 6416 6439 6450
6581 6588 6701 6716 6796 6816 6980 7156 7230 7368 7418 7446
7461 7851 8004 8179 8210 8259 8333 8438 8472 8770 8787 9031
9101 9122 9127 9168 9277 9342 9346 9428 9528 9587 9752 9794
9885 10058 10213 10233 10332 10412 10449 10487 10516 10651 10693 11002
11093 11101 11172 11190 11205 11240 11381 11439 11639 11739 11777 12109
12336 12466 12497 12576 12605 12743 12763 13074 13146 13230 13392 13395
13467 13615 13654 13686 13749 13947 14140 14376 14567 14849 14868 15036
15037 15114 15149 15223 15282 15434 15812 16018 16021 16087 16154 16232
16288 16299 16390 16715 16724 16743 16789 16843 16977 17087 17154 17373
17514 17543 17569 17709 17740 17761 17839 17858 17862 18019 18058 18099
18230 18351 18626 18710 18888 18990 19086 19330 19539 19835 19900 20074
20095 20215 20384 20389 20584 20961 20980 21125 21174 21180 21259 21312
21386 21447 21778 21909 21953 22010 22054 22190 22207 22424 22432 22433
22746 22760 22785 22967 23132 23172 23377 23383 23567 23852 23952 24000
24173 24217 24264 24306 24364 24370 24765 24950 24995 25150 25188 25215
25556 25634 25817 26167 26170 26475 26512 26515 26649 26819 26955 26957
27005 27516 27840 27997 28011 28276 28365 28369 28669 28795 28803 28933
28939 29009 29033 29075 29165 29304 29492 29516 29530 29600 30090 30112
30119 30129 30373 30462 30483 30502 30519 30600 30795 30930 31259 31337
31600 31655 31705 31852 32174 32220 32327 32402 32602 32698 32762 32808
32994 33110 33154 33160 33209 33539 33623 33691 33703 33755 33866 34215
34228 34285 34396 34510 34542 34587 34591 34610 34624 34775 34888 35292
35426 35634 35820 36027 36116 36129 36268 36576 36590 36632 36636 36788
37346 37383 37396 37418 37437 37465 37755 37822 38211 38244 38392 38429
38547 39112 39332 39758 39832 39864 39901 39951 40197 40312 40486 40572
40583 40603 40718 41689 41814 41855 41945]
Range: 41804.00
Variance: 138222933.81
Standard Deviation: 11756.82

Statistics for 'LoanAmount':
Mean: 310.45
Median: 303.00
Mode: [351. 533.]
Range: 590.00
Variance: 34010.52
Standard Deviation: 184.42
```